



COL774 / COL7341  
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$$\{x^{(i)}\}_{i=1}^m$$

$$\mu_1, \dots, \mu_K \in \mathbb{R}^n$$

$$\sum_{k=1}^K \mu_k^T \hat{\Sigma} \mu_k \quad \hookrightarrow \text{empirical co-variance matrix}$$

$$\frac{1}{m} \sum_{i=1}^m x^{(i)} x^{(i)T}$$

$\Rightarrow \mu_k = \text{eigenvector of } \hat{\Sigma}$

$$\Rightarrow \begin{cases} \hat{\Sigma} \mu_k = \lambda_k \mu_k \\ \mu_k^T \mu_k = 1 & \forall k & \text{--- (1)} \\ \mu_{k_1}^T \mu_{k_2} = 0 & \forall k_1 \neq k_2 & \text{--- (2)} \end{cases}$$

$$\mu_{k_1}^T \hat{\Sigma} \mu_{k_2} = ? \quad \frac{1}{m} \sum_{i=1}^m [\mu_{k_1}^T x^{(i)} x^{(i)T} \mu_{k_2}]$$

$$\lambda_{k_2} \mu_{k_2}^T \mu_{k_1} \quad \frac{1}{m} \sum_{i=1}^m (x^{(i)T} \mu_{k_1}) (x^{(i)T} \mu_{k_2})$$

$\Rightarrow$  How do we find eigenvectors of  $\hat{\Sigma}$

$$\hat{\Sigma} \mu = \lambda \mu \quad \begin{matrix} \text{eigenvector} \\ \text{eigenvalue} \end{matrix}$$

$$(\hat{\Sigma} - \lambda I) \mu = 0$$

$$\hookrightarrow \det(\hat{\Sigma} - \lambda I) = 0$$

SVD: Singular value decomposition

$$X \in \mathbb{R}^{m \times n}$$

$$\text{columns of } U, V = \text{orthonormal} \quad U \in \mathbb{R}^{m \times n} \quad D \in \mathbb{R}^{n \times n} \quad V \in \mathbb{R}^{n \times m}$$

$$O(n^2n, nm^2)$$

$$D = n \begin{bmatrix} \times & & & \\ & \times & & \\ & & \times & \\ & & & 0 \end{bmatrix}$$

then columns of  $V$  are eigenvectors of  $X^T X$ .

non-zero entries of  $D$  are square roots of eigenvalues of  $X^T X$

$$X = UDV^T \quad \therefore \quad X^T X = U \underbrace{D^T D}_{I} U^T \\ = U \underbrace{(D^T D)}_{n \times n} U^T$$

Post multiply by  $V$   $X^T X = V \Lambda V^T$  — (1)

$$(X^T X) V = (V \underbrace{\Lambda V^T V}_I) = V \Lambda$$

$$\Rightarrow (X^T X) v_k = v_k \lambda_k \quad \text{where } v_k \text{ is } k^{\text{th}} \text{ column of } V$$

Learning Theory:  $\{x^i, y^i\}_{i=1}^m \equiv \text{Data}$   $(x^i, y^i) \sim \text{Dist}$  i.i.d.

$h_0 \in \mathcal{H}$  :- learned  $m$   $\{x^i, y^i\}_{i=1}^m$

$$\hat{\omega}(h_0) = \frac{1}{m} \sum_{i=1}^m \frac{1}{2} h_0(x^i) \mp y^i$$

$\hookrightarrow$  training error

$(x, y) \sim \text{Dist}$

$$\omega(h_0) = E_{(x, y) \sim \text{Dist}} [\frac{1}{2} h_0(x) \mp y]$$

$$Pr \left[ \underbrace{|\hat{\omega}(h_0) - \omega(h_0)|}_{\text{PAC}} > \gamma \right] \leq \delta$$